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NOTES AND UNIQUE PHENOMENA

APPARATUS FOR DISSECTING STALKS TO EVALUATE STEM BORER INSECT RESISTANCE IN MAIZE

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Abstract

Measuring tunnel length in stalks is a common method for evaluating resistance against European corn borer (*Ostrinia nubilalis* Hübner) in maize (*Zea mays* L.). Infested plants are dissected and tunnel length, attributed to feeding of corn borer larvae, is measured. This is laborious and time-consuming, thus restricting the number of genotypes that can be evaluated for resistance in breeding programs. An apparatus built mostly of polyvinylchloride (PVC) and spring steel was constructed for dissecting maize stalks. Stalks are pushed by hand through the apparatus and are thus split apart. This apparatus dissects the stalk into two equal parts for measuring stem borer resistance in maize and reduces the time spent for splitting a stalk by ≈50% compared with conventional methods. The cutter ensures no escapes in investigating tunnel lengths.

THE EUROPEAN CORN BORER (ECB) is a major pest of maize in the U.S. Corn Belt and in maize growing regions of Canada. Research on resistance against ECB has been conducted for more than 40 yr. The area under maize cultivation in central Europe has increased dramatically during the past two decades, and this was accompanied by a continuous spread of ECB in the warmer regions, which highlights the need to screen for ECB resistance in maize breeding programs. As a common technique, tunnel length (cm of damage) in stalks is measured to evaluate second-generation ECB resistance (Guthrie and Russell, 1989; Schulz et al., 1997). The time required for dissecting maize plants and measuring tunnel length limits the number of genotypes that can be evaluated each season. Moreover, the conventional method of dissecting stalks with ordinary knives is very labor-intensive and can be a source of serious injuries. Considering these problems, we constructed an apparatus that is safe and at least twice as efficient as the conventional method for determining tunnel length from ECB infestation.

Construction and Application

Figure 1 shows the top and side view of the stalk cutter. The cutter's ground plate (240 by 90 mm), two side walls (170 by 50 mm), and two lids (90 by 80 mm, 25 by 80 mm) are made of 15-mm-thick polyvinylchloride (PVC) and are attached with 20-mm screws. The two guidance sheets (120 by 40 mm) are made of 1-mm-thick spring steel and are fixed with two screws (10 mm long) on the inner sides of the side walls. Each guidance sheet is given two 18° bends, parallel to the short edges: one bend at 20 mm from one short edge, and then another bend (still 18° but in the opposite direction) at 20 mm from the other edge. The guidance sheets constrict the feed area as the stalk approaches the knife. The knife (70 by 20 mm) has a smooth blade and can be made of an old saw-blade (0.8 mm thick). The knife, set at an angle of 65° and with the leading edge at the top, is installed in the center of the cutting unit at 110 mm from the beginning of the side walls; it is held by one screw on the back lid and one on the ground plate. The knife is removable for sharpening.

For proper use, the cutter should be fixed with a screw-clamp on a stable desk. Ears and leaves must be removed from the stalk before it is pushed through the cutter. The bottom part of the stalk is pushed by hand through the cutter first. The guidance sheets direct the stalk towards the center, where the knife is fixed, and ensure the splitting of the stalk in the middle. Because the knife is installed at an angle of 65° toward the incoming stalk, stalks are forced to stay on the ground plate during the cutting operation (Fig. 2). Finally, the bottom part of the stalk, now dissected, is pulled out by hand from the opposite side of the cutter. This completes the cutting process. Stalks with very large diameters can be split twice, to ensure that there are no escapes in investigating tunnel lengths (i.e., no hidden tunnels are missed). This construction allows cutting maize stalks of 10 to 50 mm diameter, depending on the firmness of the stalk materials (Fig. 3). For instance, stalks heavily infested with *Fusarium* spp. cannot be pushed through the cutter, because such stalks are too weak and would tear apart. Based on experience over two seasons, the procedure has proved to be safe and two times faster than the conventional method, as well as providing a more accurate centered cut and reducing escapes when

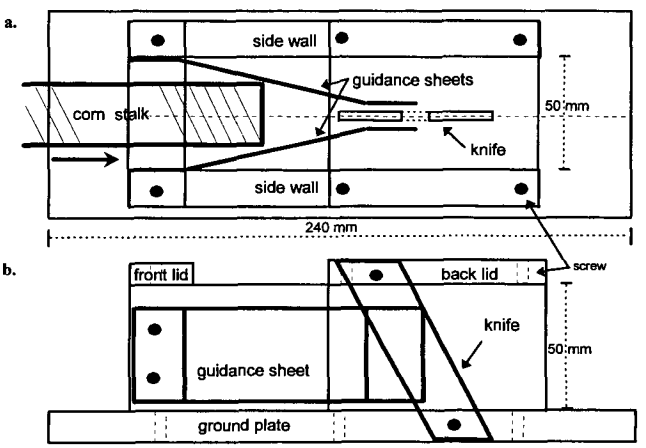

Fig. 1. Sketch of the stalk cutter: (a) top view; (b) side view. To show interior detail, the front and back lids (top view) and side walls (side view) are rendered transparent.

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Abbreviations: ECB, European corn borer.

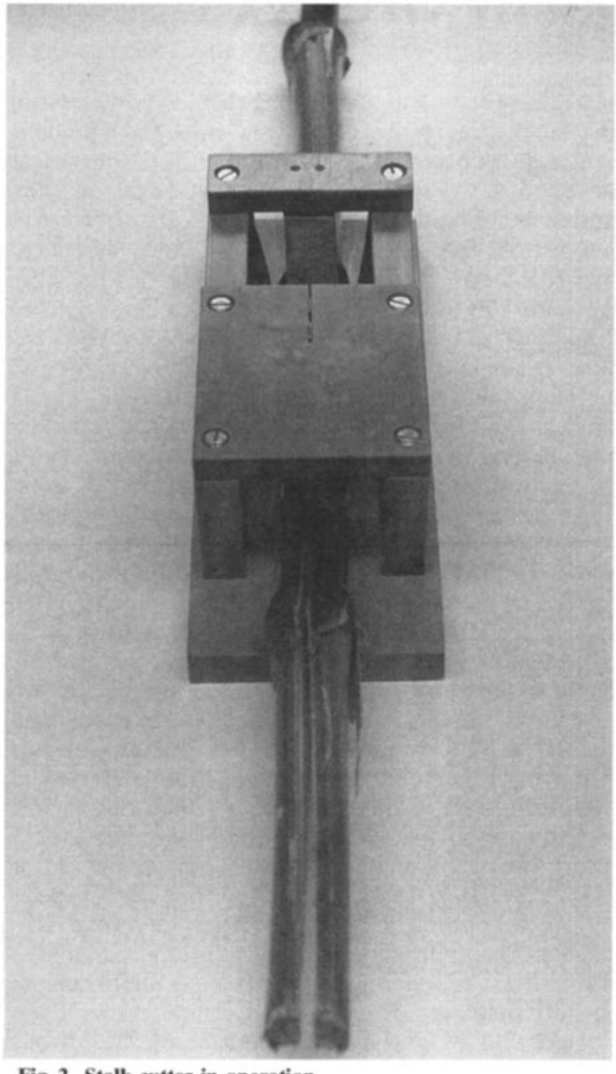

Fig. 2. Stalk cutter in operation.

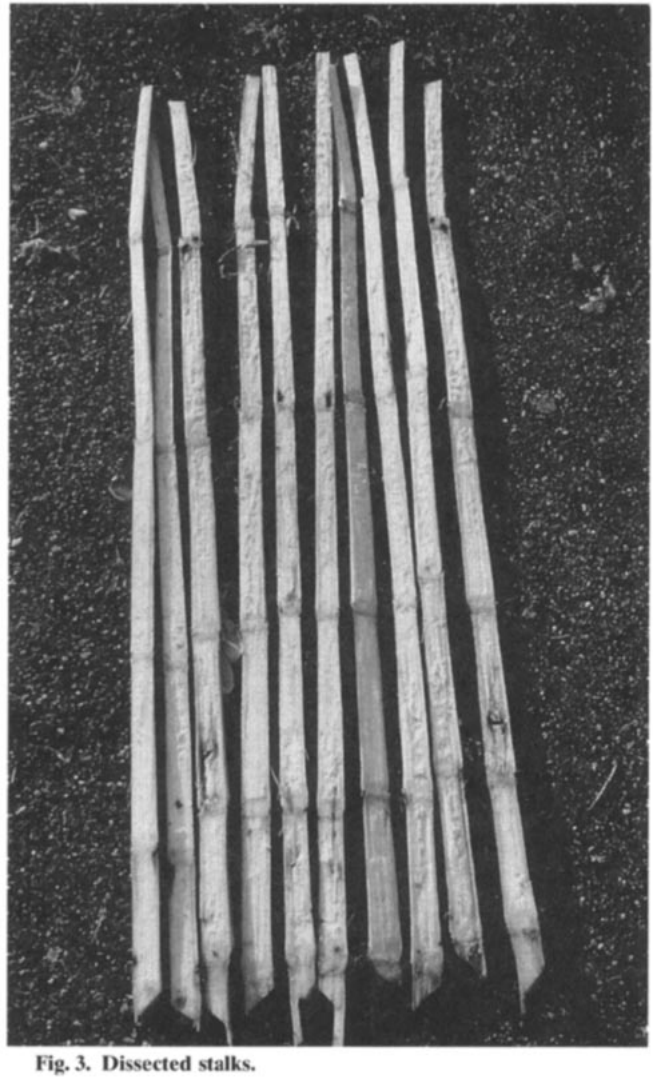

Fig. 3. Dissected stalks.

determining tunnel length. Cost of production per stalk cutter, including materials and labor, is less than $100 (U.S. dollars).

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